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$$I = \int x^9 \sqrt{x^5 + 1} dx$$

$$\text{Stel: } t = x^5 + 1 \Rightarrow dt = 5x^4 dx \Leftrightarrow dx = \frac{dt}{5x^4}$$

x^4 in de integraal ($x^9 = x^4 \cdot x^5$) valt weg samen met de x^4 bij de vervanging van dx .

$$\begin{aligned} &= \int (t-1) \sqrt{t} \frac{dt}{5} = \int (t-1) \sqrt{t} \frac{dt}{5} = \frac{1}{5} \int t^{\frac{3}{2}} - \sqrt{t} dt \\ &= \frac{1}{5} \left(\int t^{\frac{3}{2}} dt - \int \sqrt{t} dt \right) = \frac{1}{5} \left(\frac{2}{5} t^{\frac{5}{2}} + c - \frac{2}{3} t^{\frac{3}{2}} + c \right) \\ &= \frac{2}{25} t^{\frac{5}{2}} - \frac{2}{15} t^{\frac{3}{2}} + c = \frac{2(x^5+1)^{\frac{5}{2}}}{25} - \frac{2(x^5+1)^{\frac{3}{2}}}{15} + c \end{aligned}$$

References